

0, 1, 2, 3, 4, 5

pdf

Lec_6

Practice Questions on time Complexity

10

30

50
practice $\Rightarrow$
pattern 2

Loop ✓
20
0, 1, 2, 3, 4, 5

12

12
10

n

$$n = 10^{2600}$$

$$l = 12$$

$$l \leq n$$

$$l = 10^{24}$$

13.) main()

{

i=12;

while(i<=n)

{

i=i^24

}

}

$$l = 12^{240}$$

$$l = 12^{24}$$

$$l = 12^{24^2}$$

$$l = 12^{24^3}$$

$$l = 12$$

$$l = 12^{24}$$

$$l = (12^{24})^{24}$$

$$l = ((12^{24})^{24})^{24}$$

$$l \leq 10^{25}$$

$$12^{24} \leq 10^{2600}$$

$$(12^{24})^{24} \leq 10^{2600}$$

n times

$$12^{24^n} \geq n$$

=>

$$\log_{12} 12^{24^n} \geq \log_{12} n$$

=>

$$24^n \log_{12} 12 \geq \log_{12} n$$

$$\Rightarrow 24^n = \log_{12}^n$$

$$\Rightarrow \log_{24} 24^n = \log_{24} (\log_{12}^n)$$

$$\Rightarrow n \times \log_{24} = \log_{24} (\log_{12}^n)$$

$$\Rightarrow n = \log_{24} (\log_{12}^n)$$

$$O(\log_{24} (\log_{12}^n))$$

14.) main()

{

i=3;

while(i<=n)

{

i=i^2

i=i^4

i=i^5

}

}

$i = 3;$
 $\text{while}(i \leq n)$

$\{ i = i^{40}$

$\}$

14

$$3^{40^k} = n$$

$$O(\log_{40} \log_3 n)$$

$i = 12$
 $\{ i = i^{24}$

3

13

$$3^{40^k} = n$$

$$40^k \log_3 = \log_3 n$$

$$40^k = \log_3 n$$

$$n \log_{40} = \log_{40} \log_3 n$$

$i = 3^2$
 $i = (3^2)^4$
 $i = ((3^2)^4)^5$
 $i = 3^{40}$
 $i = 3$

main()

2

~~$x = x + 1$~~
 ~~$x = x + 1$~~
 ~~$x = x + 1$~~

S.

3

~~$x = x + 1$~~

~~$x = x + 1$~~

15.) main()

{

while(n>2)

{

n=n^{1/2}

}

}

$n \approx 10$

$n \approx 2$

①

②

③

n=256

256 > 2
16 > 2

4 > 2

2 > 2 X

$$n = n^{1/2} \Rightarrow n = \sqrt{n}$$

$$n = \sqrt{256} = 16$$

$$n = \sqrt{16} = 4$$

$$n = \sqrt{4} = 2$$

$$n = n^{1/2}$$

$$n = (n^{1/2})^{1/2}$$

$$n = ((n^{1/2})^{1/2})^{1/2}$$

$$= n^{1/2}$$

$$= n^{1/2^2}$$

$$= n^{1/2^3}$$

$$n = n^{\frac{1}{2^n}}$$

$$n \approx 2$$

$$n^{\frac{1}{2^n}} = 2$$

$$\log_2 n^{\frac{1}{2^n}} = \log_2 2$$

$$\Rightarrow \frac{1}{2^n} \log_2 n = 1$$

$$\Rightarrow \log_2 n = 2^n$$

$$\Rightarrow \log_2 \log_2 n = \log_2 2^n$$

$$\Rightarrow \log_2 \log_2 n = \frac{n}{\log_2 2}$$

$$\boxed{n = \log_2 \log_2 n}$$

$$n = \log_2 \log_2 2^{56}$$

$$n = \log_2 (\log_2 2^{56})$$

$$n = \log_2 56$$

$$n = \log_2 56$$

$$\boxed{n = 3}$$

16.) main()

{

i=10;

while(i<=n)

① { i=i+2

② i=i*2

③ i=i^2

}

}

$$l=10, \quad l=l+2 \\ l=12$$

$$l=l*2 \\ l=12*2 \\ l=24$$

$$l=l^2 \\ l=24^2 \\ l=(24)^2 \\ l=576$$

more changes

$$l=l+2 \Rightarrow l=(l+2)*2 \\ l=l*2 \Rightarrow l=2l+4 \\ l=l$$

$$l=(2l+4)^2 \\ l=4l^2+16+16l$$

$$l=4l^2+16+16l$$

$$l=4l^2 \\ l=l^2$$

main()
l=10
l=l^2
}

Thamms x